Fernando Daniel Martínez Cortés

Creative Developer · Frontend, WebGL & Real-Time Graphics

Guanajuato, Mexico (UTC-6, open to remote) | +52 473 124 5955 | fd.martinezcortes@ugto.mx
fernandocortes.dev | github.com/TaCodeC | linkedin.com/in/fernandocortesdev

SUMMARY

Web developer who ships production sites and real-time graphics for the browser. I work in JavaScript, Astro, Three.js and GLSL. I profile and optimize until scenes run smoothly on weak hardware, and my work on mobile VR shaders was published at PMW-2026. B.A. in Digital Arts (GPA 9.4/10), graduating in December 2026.

SKILLS

Front-End: JavaScript, HTML5, CSS3, Astro, Three.js, WebGL, GLSL, Data Visualization, Responsive Layout
Back-End & Infrastructure: CMS Integration, Linux Server Deployment, Git, Git LFS, Python
Graphics & Other: C#, C++, Java, Unity (URP), HLSL, OpenGL ES, Android SDK
Languages: Spanish (native), English (conversational)

EXPERIENCE

Technical Artist
Jun 2026 – Sep 2026
Rassone Studios | C#, Unity, HLSL
- Wrote a custom two-pass toon shader to complete the game demo.
- Built the animation and VFX setup, plus the C# editor tools behind it.
- Removed 1,832 unused components after profiling, with no change to renderer or prefab counts.

Web Developer
Jan 2025 – Aug 2026
Universidad de Guanajuato, Postgraduate Program in Biosciences (DICIVA) | Salamanca, MX
- Built and launched the program's official website, now its main public presence.
- Integrated a CMS so faculty publish news and updates without a developer.
- Deployed the site on university servers in coordination with IT.
- Organized program content into a clear structure for prospective students.

Graphics Programmer & Developer
Jan 2025 – Aug 2026
Universidad de Guanajuato × Secretaría de Salud | Salamanca, MX
- Built an AR/VR mental-health app in Unity and C#.
- Owned the shader and rendering layer under a strict mobile frame budget.
- Reduced shader instruction count and overdraw to stay within that budget.
- Co-authored the resulting paper, accepted at PMW-2026.

AR Developer, Comunidad Universitaria Scholarship
Jan 2026 – Jun 2026
Universidad de Guanajuato, Division of Engineering (DICIS) | Salamanca, MX
- Built an interactive outreach app for university career fairs, run on-site by non-technical staff.

PROJECTS

Observatorio SAMAZ, VR Observatory
github.com/TaCodeC/Observatorio-SAMAZ
Unity 6, C#, Python, Meta Quest 3
- Rendered 15,599 stars as GPU-instanced geometry with a custom shader.
- Wrote a Python pipeline to convert the star catalogue to binary.

LiveSketch, Android Drawing App
github.com/TaCodeC/LiveSketch-Android
C++, Java, OpenGL ES, NDI SDK
- Developed a multi-layer canvas that streams live over the network via NDI, used by students in class.

Portfolio & Interactive Web Work
fernandocortes.dev
Astro, JavaScript, Three.js, GLSL, HTML/CSS
- Designed Sin Fronteras, a migration-data map that uses Chernoff faces to show several variables at once.
- Created Fragment, a real-time generative visual system based on Freud's theory of the mind.
- Built and deployed this portfolio with Astro.

PUBLICATION

"Resource-Efficient Shader Implementation for Immersive Narratives on Legacy Mobile Virtual Reality Platforms"
Co-author | PMW-2026, Lviv-Kharkiv, Ukraine

EDUCATION

B.A. Digital Arts, Universidad de Guanajuato
Expected Dec 2026
Salamanca, MX | GPA 9.4 / 10.0